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MILITARY SPECIFICATION

TELEPHONE SET TA-312( )/PT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers equipment designated as Telephone Set TA-312( )/PT which is a rugged, lightweight, waterproof field telephone set.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, US Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ. 07703-5000 using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5805

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MILITARY

MIL-P-11268	Parts, Materials, and Processes Used in Electronics Equipment
MIL-G-11994	Generator, Ringing, Hand G-42( )/PT
MIL-H-11996	Handset H-60( )/PT
MIL-M-13231	Marking of Electronics Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099) *

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Electronics Command

SM-D-208032	Network, Impedance Matching CU-350( )/PT Assy
DL-SM-B-283350	Telephone Set TA-312( )/PT
SM-D-283477	Telephone Set TA-312( )/PT Test Requirements

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(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. - When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. - The equipment shall be constructed in accordance with the requirements of this specification and Drawing and Data List DL-SM-B-283350 (including the requirements for parts, materials, and processes thereon).

3.3 Parts, Materials, and Processes, General. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking

3.5.1 General. - Marking shall conform to Specification MIL-M-13231 (See 4.4). Front panel marking shall be Group 1 as described in that specification.

3.5 2 Visibility. - Wherever practical, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial Numbers. - All equipments shall have serial numbers.

3.6 Interchangeability. - Like units, assemblies, subassemblies and replacement parts shall conform to requirement 7 of MIL-STD-454 (See 4.8).

3.7 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.8 Electrical.

3.8.1 Impedance Matching Network CU-350( )/PT. -

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3.8.1.1 Input/Output. - The characteristics of the impedance matching network shall be in accordance with the input/output requirements specified in Table I of Drawing SM-D-208032. (See 4.9.1)

3.8.1.2 Dielectric Strength. - the network shall withstand 500 volts applied between all network terminals connected together and the case. (See 4.9.2)

3.8.1.3 Insulation Resistance. - the capacitors specified in Table II of Drawing SM-D-208032 shall have a minimum insulation resistance of 100 Megohms at 100 Vdc. (See 4.9.3)

3.8.2 Telephone Set TA-312( )/PT. -

3.8.2.1 DC Line Current. - (See 4.10.1)

3.8.2.1.1 Common Battery (CB). - No line current shall flow with the handset on-hook.

3.8.2.1.2 Local Battery (LB). - A line current of 55 +/- 20% ma shall flow only if the handset is off-hook and the press-to-talk switch is depressed.

3.8.2.1.3 Common Battery Signalling (CBS). - A line current of 55 +/- 20% ma shall flow with the handset off-hook.

3.8.2.2 Transmit Output. - With the equipment in the CB, LB, and CBS mode, a sound pressure level of 109 db (re .0002 dynes/cm^2) at the transmitter grid shall produce a 124 +/- 4 db (re 10^{-16} watts) line output into a 620 ohm load. (See 4.10.1)

3.8.2.3 Sidetone. - With the equipment in the CB, LB, and CBS mode, a sound pressure level of 109 db (re .0002 dynes/cm^2) at the transmitter grid shall produce a 92 +/- 6 db (re .0002 dynes/cm^2) sidetone output at the receiver. (See 4.10.1)

3.8.2.4 Generator Output. - Cranking the generator at a speed of 200 rpm shall produce a 20 Hz, 90 volt output at the line terminals. In addition, operator of the generator shall produce no output from the buzzer with the volume control in its loudest position. (See 4.10.1)

3.8.2.5 Buzzer. - An 89 +/- 5 volts, 20 Hz signal applied at the line terminals shall produce an audible signal at all settings of the volume control knob. In the minimum loudness position, the clapper ball shall not strike the diaphragm. (See 4.10.1)

3.8.2.6 Impedance. - The input and output impedance of the equipment shall be 600 +/60 ohm throughout the frequency range from 300 to 4000 Hz. (See 4.10.2)

3.8.2.7 Dielectric Strength. - The equipment shall withstand 500 volts applied between all metal parts intended to be insulated from each other. (See 4.10.3)

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3.8.2.8 Insulation Resistance. - All metal parts intended to be insulated from each other shall have a minimum insulation resistance of 100 megohms at 100 volts dc. (See 4.10.4)

3.8.2.9 Operational. - The equipment shall provide satisfactory performance in all modes of operation when tested in accordance with 4.10.5.

3.9 Fabric and Thread. - Fabric and thread shall be preshrunk or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature. (See 4.21)

3.10 Handset H-60( )/PT. - The handset shall meet all the requirements of MIL-H-11996.

3.11 Generator, Ringing, Hand G-42( )/PT. - The generator shall meet all the requirements of MIL-G-11994.

3.12 Service Conditions. - The equipment shall meet the following service conditions.

3.12.1 Temperature. -

3.12.1.1 Low. - During and after testing as specified in 4.11.1 the equipment shall meet the requirements of 3.8.2.1 thru 3.8.2.6.

3.12.1.2 High. - During and after testing as specified in 4.11.2 the equipment shall meet the requirements of 3.8.2.1 thru 3.8.2.6.

3.12.2 Humidity. - During and after testing as specified in 4.12 the equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling, or deterioration of parts and materials, and shall meet the requirements of 3.8.2.1 thru 3.8.2.6. In addition, the equipment shall meet the requirements of 3.8.2.8 after completion of the fifth cycle.

3.12.3 Altitude (Elevation). - The equipment shall show no evidence of damage or deterioration in performance at altitudes up to 10,000 feet above sea level after storage at altitudes up to 50,000 feet above sea level. (See 4.13)

3.12.4 Leakage (Immersion). - The equipment shall show no evidence of leakage after testing as specified in 4.17.

3.12.5 Vibration, Bounce, and Shock. -

3.12.5.1 Vibration.- When tested in accordance with 4.14 the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz.

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3.12.5.2 Bounce, Loose Cargo. - Following testing as specified in 4.15, the equipment shall meet the requirements of 3.8.2.9.

3.12.5.3 Shock, Drop. - Following testing as specified in 4.16, the equipment shall meet the requirements of 3.8.2.1 thru 3.8.2.6 and 3.8.2.9.

3.12.6 Fungus. - After the fungus test of 4.18, the equipment, including the canvas carrying case, shall be visually examined using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.15 square inches in area, of sparse microbial growth (See 6.3) as evidenced by growth colonization (See 6.3) which includes branching (See 6.3) and sporulation (See 6.3) on/or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (See 6.3) shall not be included in this evaluation. In addition, the equipment shall meet the electrical requirements of 3.8.2.9.

3.12.7 Salt Fog. - After the salt fog test of 4.19 the equipment, when examined visually with aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface; or exfoliation (See 6.3) of metal. In addition, the equipment shall meet the electrical requirements of 3.8.2.9.

3.12.8 Dust. - After testing as specified in 4.20 the equipment shall meet the requirements of 3.8.2.9.

3.13 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268

General requirements for plastic material and plastic parts.

Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable
- Soldering and brazing

Cleaning of Units

General Requirements for securing of parts

Process for welding

Cleaning after welding

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Controls

Self-tapping screws

Cleaning prior to welding

3.14 System Safety Engineering

3.14.1 Personnel Hazards. - Personnel hazard risk levels shall not be increased over those established for the development models or production system (equipment). No manufacturing process, selection of parts, or product configuration changes shall cause or contribute to the creation of a personnel hazard. The produced item shall conform to the safety design criteria of Requirements 1 and 8 of MIL-STD-454K, Standard General Requirements for Electronic Equipment, and paragraph 5.13 of MIL-STD-1472C, Hazards and Safety.

3.14.2 Radioactive Materials. - Radioactive materials will not be used without full approval of the Government. Any use of radioactive materials shall conform to all Nuclear Regulatory Commission (NRC) and Army regulations to assure minimum hazard to personnel during manufacture, use, transportation, storage, and disposal.

3.14.3 Visual and Mechanical Inspection. - An inspection shall be performed on a typical production unit utilizing SEL Form 1183, System Safety Design Verification Checklist, to verify compliance of the manufactured item with the design specifications or procurement requirements that apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. - All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of Inspections. - The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3). Does not include packaging.

b. Inspections covered by subsidiary documents (See 4.4).

c. Quality conformance inspections:

(1) Quality conformance inspection of equipment before packaging.

4.2.1 Inspection conditions. - Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraph(s) in the specification.

4.3 First Article. - Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article Units. - The contractor shall furnish the number of First Article units in accordance with the following:

a. Four (4) each Telephone Sets TA-312( )/PT

b. Four (4) each Generator, Ringing, Hand G-42( )/PT

c. Four (4) each Handset H-60( )/PT

d. Four (4) each Network CU-350( )/PT

4.3.2 First Article Inspection. - The First Article inspection shall consist of the inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First Article Data. - The First Article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspections Covered by Subsidiary Documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

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ITEM	Where Required
Construction	3.2
Parts, Materials and Processes	3.3
Finish	3.4
Marking	3.5
Handset H-60( )/PT	3.10
Generator G-42( )/PT	3.11

TABLE I. - First Article Inspection

Insp	Reqmt	Test	Units 1 & 2	Units 3 &4
Insp. covered by subsidiary document		4.4	All Units	
Bounce Preconditioning	3.7	4.6	All Units	
Group A Inspection	See Table II		All Units	
Group B Inspection	See Table III		All Units	
Environmental				
Low Temp	3.12.1.1	4.11.1	1	2
High Temp	3.12.1.2	4.11.2	2	3
Humidity	3.12.2	4.12	3	4
Altitude	3.12.3	4.13		1
Vibration	3.12.5.1	4.14	4	
Bounce, Loose Cargo	3.12.5.2	4.15	5	
Shock, Drop	3.12.5.3	4.16	6	
Fabric Shrinkage	3.9	4.21		5
Fungus	3.12.6	4.18	7	
Dust	3.12.8	4.20		6
Salt Fog	3.12.7	4.19		7

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. - The contractor shall perform the inspections specified in 4.4, 4.6, and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A Inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection.

4.5.1.1 Order of Inspection Within Group A. - Group A inspection shall be performed in an order satisfactory to the government, except that the operational in 4.10.5 shall be last.

TABLE II. - Group A Inspection

Inspection	Reqmt	Insp
Visual Mechanical	3.13	4.7
Electrical		
CU-350( )/PT		
Input/Output	3.8.1.1	4.9.1
Dielectric strength	3.8.1.2	4.9.2
Insulation resistance	3.8.1.3	4.9.3
TA-312( )/PT		
D.C. Line current	3.8.2.1	4.10.1
Transmit output	3.8.2.2	4.10.1
Sidetone	3.8.2.3	4.10.1
Generator output	3.8.2.4	4.10.1
Buzzer	3.8.2.5	4.10.1
Impedance	3.8.2.6	4.10.2
Operational	3.8.2.9	4.10.5

4.5.2 Group B Inspection. - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III.

4.5.2.1 Order of Inspection within group B. - Group B inspection shall be performed in an order satisfactory to the government.

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TABLE III. - Group B Inspection

Inspection	Reqmt	Test
Interchangeability	3.6	4.8
Leakage (Immersion)	3.12.4	4.17
Dielectric strength	3.8.2.7	4.10.3
Insulation resistance	3.8.2.8	4.10.4

4.5.3 Group C Inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C Inspection. -

4.5.3.1.1 Subgroup I. - For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every 3 months or 6000 units or fraction thereof produced, whichever occurs first.

4.5.3.1.2 Subgroup II. - For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every 7 months.

4.5.3.2 Order of Inspection within Group C. - Group C inspection shall be performed in the order specified in Table IV.

4.5.3.3 Group C Failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass the leakage (immersion) test of 4.17 and Group A inspection, in that order.

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TABLE IV. - Group C Inspection

Insp	Reqmt	Test	Unit 1	Unit 2
Subgroup I:				
Low Temp	3.12.1.1	4.11.1	1	2
High Temp	3.12.1.2	4.11.2	2	3
Humidity	3.12.2	4.12	3	4
Altitude	3.12.3	4.13		1
Vibration	3.12.5.1	4.14	4	
Bounce, Loose C.	3.12.5.2	4.15	5	
Shock, Drop	3.12.5.3	4.16	6	
Fabric	3.9	4.21		5
Subgroup II:				
Fungus	3.12.6	4.18		1
Dust	3.12.8	4.20	1	
Salt Fog	3.12.7	4.19	2	

4.5.4.1 Noncompliance. - If a sample fails to pass Group C or D inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, and on all units of production which were manufactured with the same materials and processes, and which are subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action and acceptance to the qualifying activity has been taken. After the corrective action has been taken, Group C or D inspection shall be repeated on additional samples (all tests and examinations , or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstitute; however, final acceptance and shipment shall be withheld until the Group C or D inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6 Preconditioning Bounce. - The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

4.7 Visual and Mechanical Inspection of Equipment. - Equipment shall be examined for the defects listed in MIL-STD-252.

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4.8 Inspection for Dimensional Interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability of requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.9 Electrical Tests for CU-350( )/PT. -

4.9.1 Output Voltage. - The network shall be tested in accordance with SM-D-208032.

4.9.2 Dielectric Strength. - The network shall be tested in accordance with Method 301 of MIL-STD-202. A test voltage of 500 volts shall be applied for at least 30 seconds between all network terminals connected together and the case.

4.9.3 Insulation Resistance. - The network shall be tested in accordance with test condition A, Method 302 of MIL-STD-202. Measurements shall be made across terminals 1 & 5 and across 5 & 6.

4.10 Electrical Tests for TA-312( )/PT. -

4.10.1 Input/Output. - The equipment shall be tested in accordance with Drawing SM-D-283477.

4.10.2 Impedance. -

4.10.2.1 Output. - Connect the equipment as shown in Figure 1. Set the oscillator to 300 Hz, .4 volt on the VTVM. Close SW1 and vary the precision decade resistor until the VTVM reads .2 volt. Record the value of the decade resistor for compliance with 3.8.2.6. Repeat the procedure with the oscillator set to 1kHz, 2kHz, 3kHz, and 4kHz.

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4.10.2.2 Input. - Connect the equipment as shown in Figure 2, with the decade resistor set to 0 ohm. Adjust the d.c. power supply for 50 ma on M1. With SW1 in position A, set the audio oscillator to 300 Hz, 0.4 volt on the VTVM. Move SW1 to position B and vary the precision decade resistor until the VTVM reads 0.2 volt. Record the value of the decade resistor for compliance with 3.8.2.6. Repeat the procedure with the oscillator set at 1kHz, 2kHz, 3kHz, and 4kHz.

4.10.3 Dielectric Strength. - The equipment shall be tested in accordance with Method 301 of MIL-STD-202. A test voltage of 500 volts RMS shall be applied for at least 10 seconds as follows:

a. From each line terminal post to case

b. From each battery contact to case

c. From each external battery contact to case

d. Across the external battery contacts

e. Across pins L & J of U-79/U receptacle

f. Across pins F & H of U-79/U receptacle

g. From each pin of U-79/U receptacle to case

4.10.4 Insulation Resistance. - The equipment shall be tested in accordance with test condition A, Method 302 of MIL-STD-202. Measurements shall be made at the points specified in 4.10.3.

4.10.5 Operational. -

4.10.5.1 CB Circuit. - With selection switch set to CB, a clicking sound shall be heard when the generator is cranked slowly; sidetones shall be noted when generator is cranked at high speed (200 rpm); sidetones shall be eliminated and the torque of the generator crank shall be reduced by depressing the hook switch while cranking generator at 200 rpm.

4.10.5.2 LB Circuit. - With selector switch set to LB and two BA-30 batteries in place, sidetones shall be noted with press-to-talk switch depressed; sidetones shall be eliminated when press-to-talk switch is released; torque of generator shall be increased when press-to-talk switch is depressed.

4.10.5.3 CBS Circuit. - With the selector switch set to CBS and two BA-30 batteries in place the telephone shall perform as in 4.10.3.2 except the torque of the generator shall be independent of the press-to-talk switch.

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4.10.5.4 Buzzer Circuit. - With two TA-312( )/PT telephone sets connected together the cranking of the generator on one set shall cause the buzzer on the other set to operate. Rotate the volume control knob and note the change from Low to Loud. The buzzer shall be audible at all settings of the volume control knob.

4.10.5.5 Overall Test. - Connect two TA-312( )/PT telephones with batteries in place, INT-EXT switch set to INT, and the mode selector switch set to LB or CBS. A two-way conversation shall be established between the two sets. On one telephone set the INT-EXT switch to EXT and connect an external handset to the U-79/U receptacle. Remove the batteries on the other telephone and connect a 3 volt power source to the BAT terminals. A two-way conversation shall be established between the two sets.

4.11 Temperature Test. -

4.11.1 Low Temperature. - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -80°F for a period of 24 hours, and the operating temperature shall be -40°F.

4.11.2 High Temperature. - The equipment shall be tested in accordance with Method 501, Procedure I, of MIL-STD-810. The operating temperature shall be 125°F.

4.12 Humidity Test. - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Measurements shall be made during the periods shown on the figure for the cycle.

4.13 Altitude Test. - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. The equipment shall be operated in accordance with 4.10.5.5, except external power source and handset are not required.

4.14 Vibration, Resonance Search Test. - The equipment shall be tested in accordance with Method 514.1, Procedure IX, Part I of MIL-STD-810.

4.15 Bounce, Loose Cargo Test. - The equipment shall be tested in accordance with Method 514.1, Procedure XI, Part 2 of MIL-STD-810.

4.16 Shock, Drop Test. - The equipment shall be tested in accordance with Method 516.1, Procedure II of MIL-STD-810.

4.17 Leakage (Immersion) Test. - The equipment, without carrying case, shall be immersed in fresh water to a minimum depth of 3 feet for 10 minutes. The equipment and the water shall be at ambient room temperature. After 10 minutes the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage except 2 or 3 drops at the ceramic breathers.

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4.18 Fungus Test. - The equipment shall be subjected to the fungus test specified in Method 508, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (See 6.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during test, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.12.6 shall constitute failure of this test.

4.19 Salt Fog. - The equipment shall be subjected to the salt fog test specified in Method 509, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the equipment to meet the requirements of 3.12.7 shall constitute failure of this test.

4.20 Dust Test. - The equipment shall be tested in accordance with Method 510, Procedure I, of MIL-STD-810.

4.21 Test for Shrinkage. - the carrying case shall be inspected for conformance to 3.9.

5. PACKAGING

5.1 Packaging requirements. - The packaging requirements for the desired level(s) or protection shall be as specified by the acquisition activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. - Telephone Set TA-312( )/PT is intended for use as a local or common battery telephone set for fixed, mobile, and point-to-point communication. The telephone set is applicable for outdoor use in combat areas and for general use as a desk or wall instrument.

6.2 Acquisition requirements. - Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Levels of preservation and packing (see section 5).

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d. When the rough handling test is required.

e. Marking and shipping of samples.

f. Place of the final inspection.

6.3 Definitions. -

6.3.1 Branching. - Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.3.2 Exfoliation. - Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.3.3 Growth Colonization. - Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.4 Microbial Growth. - Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eyes.

6.3.5 Sporulation. - Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.3.6 Tubular Germination. - Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4 Group C Inspection. - Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action (See 4.5.3.3).

6.5 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TA-312W/PT. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (See 1.1).

6.6 Verification Inspection. - Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.7 Location of Operational Inspection. - It is desirable that the operational inspection (4.10.5) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Subject term (keyword) listing. -

DC Line Current
Common Battery (CB)
Local Battery (LB)
Common Battery Signalling
Sidetone
Buzzer
Impedance
Dielectric Strength
CB Circuit
LB Circuit

6.9 Changes from previous issue. - Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - (CR)

Navy - YD

Air Force - 99

Review Activities:

Air Force - 80

User activities:

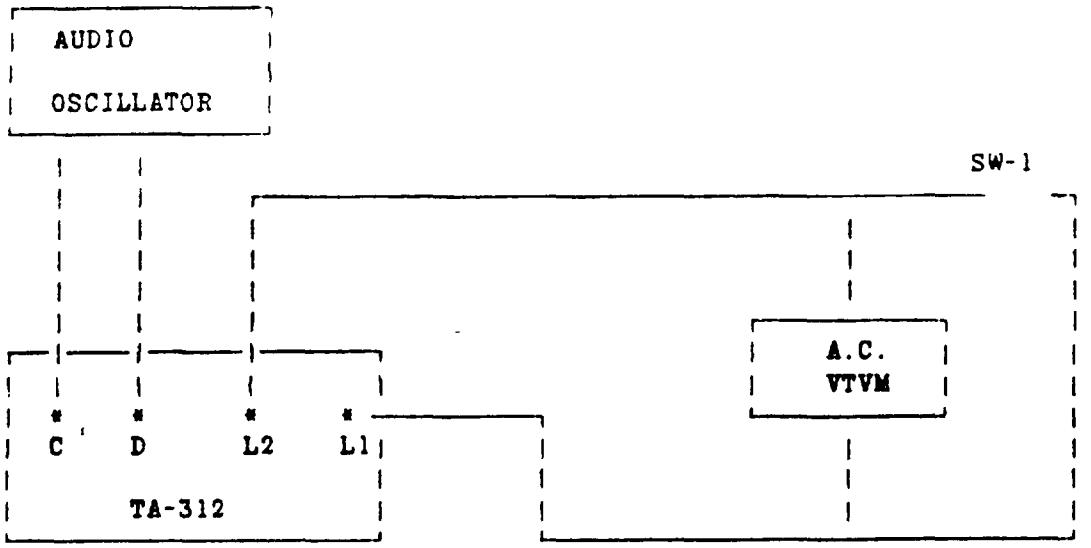
Navy - MC

Preparing Activity

Army - CR

Project No. 5805-0045

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TA-312( )/PT SWITCH POSITIONS

INT EXT - EXT

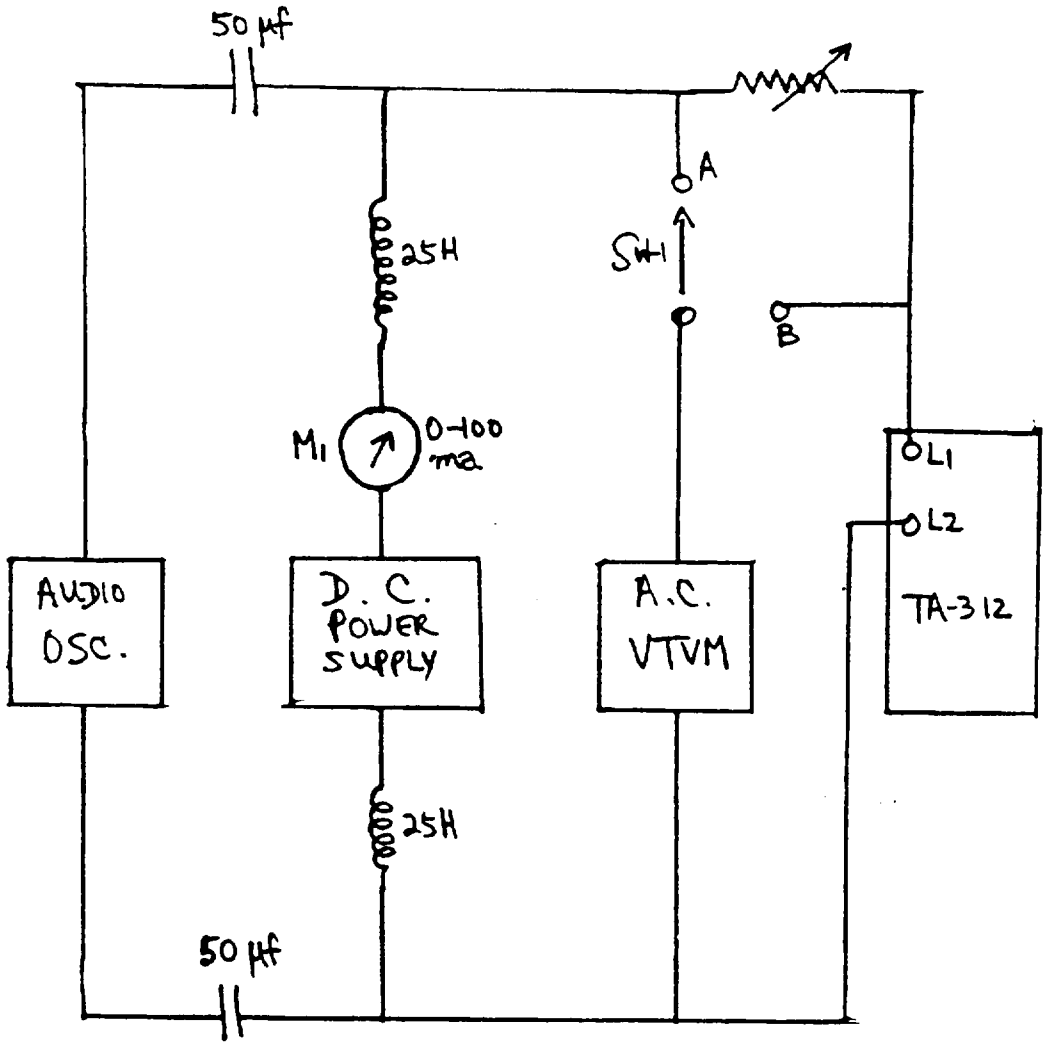
CB LB CBS - CB

NOTES:

1. AUDIO OSCILLATOR IS CONNECTED TO PINS C & D OF THE U-79/U RECEPTACLE

FIGURE 1. OUTPUT IMPEDANCE

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TA-312( )/PT SWITCH POSITIONS

INT	-	EXT	-	INT
CB	LB	CBS	-	CB

FIGURE 2. INPUT IMPEDANCE

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Fort Monmouth, NJ 07703-9990

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-14358C	Telephone Set TA-312( )/PT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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